

Various DSL Technologies And How They Differ

This article explores the information carrying capability of different types of DSL technologies along with the limitations of various existing high-speed data services like coaxial cable, analogue modem, cable modem, ISDN and satellite data services



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The rapid development in the field of information technologies has led to the appearance of new services that require high-speed data transmission technologies. For example, services like voice-over-IP, video streaming, teleworking, telemedicine, telecommuting, broadcasting of TV programmes, high-speed file sharing, user-generated video, online video gaming, online education, shopping etc, require high-speed Internet access for effective operation.

Until now, existing high-speed services (coaxial cable, analogue modem, etc) were not well-suited to the real needs of these services due to several

different reasons. The idea of using twisted-pair cabling seemed the best since throughout the world millions of connections of this type were already in place and it just needed equipment to be added to the telephone exchange along with a small installation at the user location to be able to access digital subscriber line (DSL) technology. Unshielded copper pairs used in the telephone network carry voice signals in the frequency range of 300 to 3400 Hz but are capable of transporting information at much higher rates. These cables have been used to transport data in LANs up to 10 MHz or more.

The performance of analogue modems is very poor in comparison to the DSL. Another solution lies in modem

bonding, where two modems are close-coupled to one computer, which may theoretically double the performance of a single modem. Theoretically, this arrangement increases network performance by a factor of two, but many Internet users need much more improvement in their connectivity speeds to support features like video streaming, online gaming and large file transfer at the home or workplace. For these purposes, DSL is a better option.

Cable modems support high-speed, 'always-on' Internet access using the cable television lines, which is comparable to that of DSL. But, the difference between DSL and cable line technology lies in the distribution of bandwidth. In DSL the bandwidth is dedicated locally to all the subscribers, i.e., speed won't drop when others use it at the same time. On the other hand, cable modem service involves locally shared bandwidth. This means, the realised performance of a customer's cable will depend on how many other customers in that local area subscribe to the same service.

Integrated services digital network (ISDN) technology provides data rates just twice that of the ordinary dial-up connection. But this much speed is not comparable to data rates of either cable modem technology or DSL technology. ISDN has been more widely available for several years from the telecom companies, but the very fast expansion of DSL networks superseded the ISDN.

For remote areas, which are out of reach of DSL service, satellite data service is a good option. But, here again, data rates are nearly one-third the data rates of DSL.